SUPPORTING ADULT GUIDANCE

Shapes that hold their form

Constructions, Loci and Bearings · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson moves from constructing single triangles (Lesson 1) to constructing quadrilaterals and, crucially, to the idea of **rigidity**. Learners discover that three side lengths fix a triangle (SSS) so it cannot be deformed, whereas a quadrilateral with fixed sides retains one degree of freedom and can flex — until a diagonal triangulates it. The dynamic interactives let learners feel this directly by dragging fixed-length frames.

Driving Question: *Which shapes are fixed by their sides alone?*

The mathematics, in plain terms

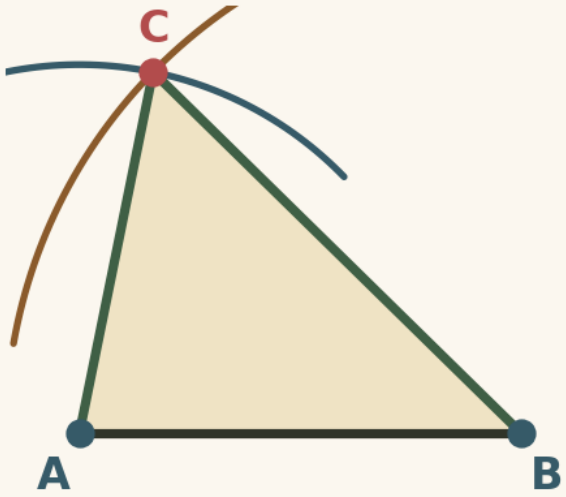
A triangle has three sides and three angles, but once the three sides are set the angles are forced too — so the shape is completely determined (this is the SSS congruence idea). A quadrilateral has four sides and four angles; fixing the four side lengths still leaves one angle free, so it can swing into different shapes. Adding a diagonal creates two triangles that share that diagonal; since each is rigid, the quadrilateral can no longer move. Engineers call this triangulation and rely on it in almost every rigid structure.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Construct a triangle from three sides

Construct a triangle with sides 6 cm, 5 cm and 7 cm — the single shape those three sides fix.

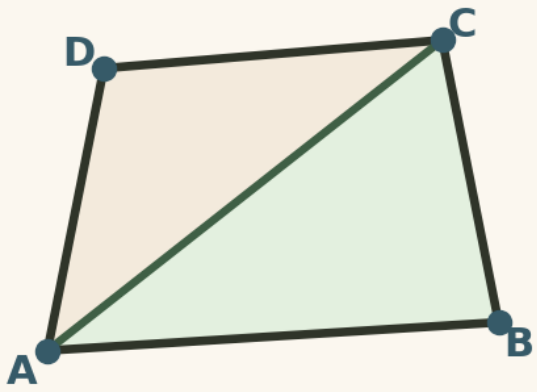


1. Draw the base AB = 6 cm.	AB = 6 cm
2. Swing an arc from A, radius 5 cm.	radius = 5 cm
3. Swing an arc from B, radius 7 cm.	radius = 7 cm
4. Mark C where the arcs cross; join A–C and B–C.	join A–C, B–C

There is only one shape of triangle with sides 6, 5, 7 (SSS): the three sides fix it completely.

Example 2 — Brace a quadrilateral to make it rigid

A quadrilateral ABCD can flex even with fixed sides. Add one diagonal to make it rigid.



1. Construct the quadrilateral ABCD (its four sides).	4 sides
2. Draw the diagonal from A to C.	join A–C
3. The diagonal splits ABCD into triangles ABC and ACD.	2 triangles
4. Each triangle is rigid, so the whole frame is now rigid.	rigid

One diagonal triangulates the quadrilateral: two rigid triangles lock the shape.

Feeling rigidity, not just being told it

The temptation is to state 'triangles are rigid' as a fact to memorise. The lesson instead lets learners drag fixed-length frames and *watch* a triangle refuse to change while a quadrilateral flexes. Encourage the learner to narrate what they see — 'the sides are the same length but the shape changed' for the quadrilateral, and 'it won't move at all' for the triangle. The construction work then makes the shapes for real.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Meet rigidity in the world — spot the triangles engineers build into a bridge truss.	~9 min

Movement	Drag two fixed-length frames and discover a triangle is rigid while a quadrilateral flexes.	~14 min
Reflection	Explain why three fixed sides fix a triangle's angles, while a quadrilateral keeps one free.	~9 min
Creativity	Add a diagonal to triangulate a quadrilateral and watch it lock rigid.	~9 min
Rest	A short pause to let the sturdy idea settle.	~2 min
Nutrition	Name what the learning feeds: understanding why, and seeing geometry hold up real structures.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

“A quadrilateral with equal sides must be a fixed shape.”

Not so — a rhombus can flex into many shapes with the same four sides. Ask the learner to drag the quadrilateral frame and watch the sides stay equal while the shape changes.

“The triangle is rigid because it is smaller / stronger material.”

Rigidity is about freedom, not material or size. Ask: 'Once the three side lengths are fixed, is there anything left that could change?'

“Any extra line inside will make a shape rigid.”

It must be a diagonal that creates triangles. Ask the learner to check that their brace actually splits the quadrilateral into triangles.

Erasing construction arcs to tidy up.

The arcs are the evidence the triangle was constructed, not estimated. Ask them to keep the arcs and point to the one that fixes each side.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: constructing with compass arcs (Lesson 1), and what a triangle and quadrilateral are. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner insists the quadrilateral 'should' be fixed because its sides do not change, or is unsure why a diagonal helps. The aim is a question that returns the thinking to them, not a lecture.

- “The four sides stayed the same length when you dragged it — so what changed?”
- “How much freedom is left once a triangle's three sides are fixed? And once a quadrilateral's four are?”
- “When you add the diagonal, what two shapes appear inside the quadrilateral?”
- “Why would each of those two shapes refuse to change on its own?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Do we need a protractor for this lesson?

No. Everything is built with compass arcs and a straightedge; angles are observed, not measured. A protractor comes later in the unit.

Is 'rigid' a real mathematical idea or just engineering?

Both. Mathematically it is the SSS congruence result — three sides determine a unique triangle — which is exactly why engineers triangulate structures.

My learner built a quadrilateral that would not flex. What happened?

They may have already added a diagonal, or fixed a corner. On the screen frame, only the four sides are fixed; if it will not move, check whether a diagonal is present.

What should I look for in good work?

A triangle and a quadrilateral both constructed (arcs showing), a diagonal added to the quadrilateral, and a sentence explaining why the triangle is rigid and the plain quadrilateral is not.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like

Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.